

## Test Planner and Tracker - Team 28

### Team Fleetix

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| Test ID | Related Use case               | Pre-condition(s)                                                                                                        | Test Description                                                                                                                                                                                      | Expected Outcome                                                                                                                                                                                                                                                              | R1 Outcome | Comments (if test case failed) | R2 Outcome | Comments (if test case failed)                                                            |
|---------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------------|------------|-------------------------------------------------------------------------------------------|
| TC-001  | UC001 - Geocoding              | System is operational. Address strings are available for processing and are recognizable by the OSM database.           | 1. Enter multiple address strings into the system<br>2. System processes these addresses through the Geocoding API<br>3. Verify coordinates are returned                                              | Returns valid geographic coordinates (latitude, longitude), that are accurate (within ~500 meters of the real-world location).                                                                                                                                                |            | NA                             |            | NA                                                                                        |
| TC-002  | UC004 - Routing (Many-to-One)  | System is operational. Source and destination coordinates are valid.                                                    | 1. Enter a valid source address<br>2. Enter a valid destination address<br>3. Request route generation<br>4. Verify route details                                                                     | A route is generated along with details like total distance (in km), estimated duration (based on historic data). The route must follow real-world roads and minimise time.                                                                                                   |            | NA                             |            | NA                                                                                        |
| TC-003  | UC004 - Routing (Many-to-One)  | System is operational. Source, destination, and multiple pickup coordinates are valid.                                  | 1. Enter a valid source address<br>2. Enter a valid destination address<br>3. Enter multiple pickup locations<br>4. Request route generation<br>5. Verify route details and ordering                  | Along with the expectations from TC-002, this use case should return the ordered lists of the waypoints that minimises the overall time taken.                                                                                                                                |            | NA                             |            | NA                                                                                        |
| TC-004  | UC008 - Routing (Many-to-Many) | System is operational. Multiple source and destination coordinates are valid. Multiple agents (vehicles) are available. | 1. Define multiple pickup locations<br>2. Define multiple destination locations<br>3. Specify number of available vehicles<br>4. Request route optimization<br>5. Verify route assignments and timing | Along with the expectations from TC-002, this use case should return the ordered lists of the waypoints that minimises the overall time taken for each agent. The distribution between the agents should be such that the time taken by the maximum-timed agent is minimised. |            | NA                             |            | NA                                                                                        |
| TC-005  | UC005 - Visualise Route        | System is operational. Routes have been generated and are available for visualization.                                  | 1. Verify visual elements<br>2. Test map interaction features.                                                                                                                                        | Route is displayed on a map with markers to clearly signify source, destination, and pickups. Visuals must include colour-coded (in case of many-to-many) path overlay Color coding per agent. Also provide map interaction like zoom, pan, move around, etc                  |            | NA                             |            | NA                                                                                        |
| TC-006  | UC009 - User Interface         | System is operational.                                                                                                  | Intuitive application design with optimised user interface for improved and efficient user experience                                                                                                 | Accessed the system through web interface<br>Navigated through main sections<br>Verified responsiveness and functionality                                                                                                                                                     |            | NA                             |            | Work in Progress. User Interface is under updation (due to integration of newer features) |
| TC-007  | UC009 - User Interface         | System is operational. Address input component is available.                                                            | 1. Begin typing an address in the address input field<br>2. Observe system suggestions<br>3. Time the response<br>4. Select a suggestion and verify autofill                                          | System returns a list of suggested complete address options ordered by relevance. Dynamically updates as the user types and autofills when user selects. Fast response time of <0.5s                                                                                          |            | NA                             |            | NA                                                                                        |

|        |                                                         |                                                                                       |                                                                                                                                                                                                                                     |                                                                                                                                                             |  |                       |  |                                                            |
|--------|---------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------|--|------------------------------------------------------------|
| TC-008 | UC009 - User Interface                                  | System is operational. User management module is implemented.                         | 1. Create accounts for each user role (admin, driver, employee)<br>2. Attempt to access role-specific features with each account<br>3. Verify appropriate access controls                                                           | Each user role has access to their appropriate interface and features. Admins can manage users, drivers can view routes, and employees can set preferences. |  | Not implemented in R1 |  | Implementation in progress. Testing yet to be performed.   |
| TC-009 | UC003 - Employee Pick-up & Drop-off Location Preference | System is operational. Employee accounts are created.                                 | 1. Log in as an employee<br>2. Navigate to preferences section<br>3. Set preferred pickup location<br>4. Set preferred drop-off location.<br>5. Save preferences<br>6. Generate a route that includes this employee                 | Employee can successfully set and update pickup/drop-off preferences. These preferences are saved to their account and used when generating routes.         |  | Not implemented in R1 |  | Implementation in progress. Testing yet to be performed.   |
| TC-010 | UC006 - Adding Constraints                              | System is operational. Admin account is created.                                      | 1. Log in as admin<br>2. Navigate to constraint management<br>3. Add various constraints (vehicle capacity, time windows, maximum route duration)<br>4. Generate routes with these constraints<br>5. Verify constraints are applied | Admin can successfully add and manage various constraints. These constraints are properly applied when generating routes.                                   |  | Not implemented in R1 |  | Maximum route duration constraint not properly implemented |
| TC-011 | UC007 - Ride History and Reports                        | System is operational. Previous rides data is available.                              | 1. Log in as admin or employee<br>2. Navigate to ride history section<br>3. View past rides<br>4. Filter rides by date, location<br>5. Generate reports                                                                             | User can view past rides with details including route, timing. Filtering options work correctly and reports can be generated.                               |  | Not implemented in R1 |  | Yet to start with its implementation.                      |
| TC-012 | UC009 - User Interface                                  | System is operational. Admin can add, edit and remove an employee or a driver details | 1. Log in as admin<br>2. Navigate to user management<br>3. Perform the CRUD operation<br>5. Verify all changes are applied throughout                                                                                               | Admin can successfully add and manage all the non-admin users in the database.                                                                              |  | Not implemented in R1 |  | Yet to start with its implementation.                      |

#### Key:

All tests passed with expected results

Implementation / Updates in progress.

Yet to be implemented

At least one test failing or not matching expected output. Needs resolve

#### Note:

1 By optimised routes, we mean that the time taken on the route is at par with the one suggested by Google Maps (SOTA mapping and navigation service), with an admissible deviation of max(5mins, 10% of total OSRM route predicted route time)

2 While we have marked Geolocation as Green (All tests passed with expected results), the OSM data is inherently weak in extracting geo-codes from any address string. For instance, "BITS Hyderabad", "BITS Pilani, Hyderabad Campus", or even "Birla Institute of Technology & Science Pilani, Hyderabad Campus" (official / google term) yields no result. It needs the exact term "Birla Institute of Technology & Science Hyderabad" to get the desired result.